COMP 018: Database Administration — Activity 6 Progress Report

Database Backup Planning, Verification, Restoration, and Disaster Recovery Simulation

Project Identification & Metadata

Course	COMP 018 - Database Administration
Evaluating Group	Group 4: Olivarez, Abangan, Pocsidio, Infantado, and Bartolata
Evaluating Group Repository	https://github.com/Nycri/brewpos
Target Project Monitored	Campus Canteen Pre-Order and Queue Management System
Target Project Group	Group 8 (bwisIT Solutions): Joshua Borreta, Martin Lawrence Bola, Joaquin Phillip Juatchon, Karl Raphael Marcos, Sancho Pacheco Jr, Euan Joseph Vera Cruz
Target Project Repository	https://github.com/EclipseChan/bwisIT-Solutions---Campus -Canteen-Pre-Order-and-Queue-Management-System
Primary Database	canteen_db (PostgreSQL 18.6 on x86_64 Windows, compiled by MSVC)
Restored Test Database	canteen_db_test_restore

ACTIVITY 6: Database Backup, Restoration, and Recovery

3.1 Backup Planning & Architecture

The disaster recovery plan for the Campus Canteen database encompasses seven relational tables within the public schema, covering student accounts, active stall menus, live order transactions, queue numbers, and audit notifications. The plan captures sequence counters, primary keys, foreign keys, constraints, and privilege grants to ensure operational continuity.

The backup strategy utilizes PostgreSQL custom compressed archive format. Dump files are stored in a protected directory outside web roots and repository trees. Recovery objectives define a Recovery Point Objective of 15 minutes and a Recovery Time Objective of 30 minutes.
Retention guidelines specify maintaining daily backups for seven days, weekly archives for four weeks, and monthly snapshots for one term.

Because database login accounts exist at the PostgreSQL cluster level and are not stored within single-database dumps, disaster recovery plans require executing access-control role scripts on new server instances prior to restoring database contents.

3.2 Backup Creation, Checksum & Archive Inspection

Logical backup creation was executed using administrative backup utilities, producing a compressed custom format archive sized at 27,405 bytes. Cryptographic verification computed a SHA-256 hash value of 1F94752EAEFA683AFF69E12554130D8D872C0245EB6A4F46D4CB27B07D36988F, establishing a baseline for integrity verification.

Archive table of contents inspection verified the inclusion of all seven database tables, sequence objects, primary key indexes, and foreign key constraint definitions. To ensure recoverability beyond file inspection checks, a complete restoration drill was conducted in an isolated test database.

3.3 Restoration into Separate Test Database

An isolated test database named canteen_db_test_restore was created on the local server instance. The custom archive file was restored into this database using administrative utility tools. Schema usage permissions were adapted to grant necessary access rights to table owners, allowing the restore process to complete cleanly in 0.41 seconds.

3.4 Post-Restore Verification Suite

Post-restoration verification confirmed that all seven relational tables were recreated with proper schema structure under administrative ownership. Record count validation across tables demonstrated complete parity with the source database, matching four student records, five vendor stalls, twelve menu items, four customer orders, five order items, four queue entries, and ten notification logs. All 58 database constraints were verified active.

Access control verification on the restored test database confirmed that privilege definitions transferred accurately. Queries executed under the application runtime account successfully

returned vendor listings, while multi-table business queries returned order statuses and queue numbers correctly.

3.5 Controlled Recovery Simulation

A controlled data recovery simulation was performed to test incident response procedures. A disposable menu item record was inserted into the restored test database to establish a baseline state. A data loss event was then simulated by removing the record, and queries confirmed its deletion.

Targeted record restoration was executed to reinsert the missing data row without affecting existing database entries. Post-recovery verification confirmed that the record was fully restored with its original attributes and foreign key relationships intact.

3.6 Limitations, Evaluation & Production Recommendations

Post-restore permission adaptation demonstrated that restoring into newly created databases requires granting explicit schema usage permissions to table owners following public privilege revocation. For production operations, daily automated backup generation should be scheduled using system task schedulers to archive database dumps with date-stamped filenames. Enabling continuous write-ahead log archiving will support point-in-time recovery capabilities to meet sub-15-minute recovery objectives, while monthly restore drills ensure backup file integrity.

CLASS GOOGLE SHEETS TRACKER ENTRY (Activity 6)

Column Header	Value
Project	Group 8 - bwisIT Solutions: Campus Canteen Pre-Order and Queue Management System
Database and backup file	canteen_db, canteen_db_activity6.backup, Size: 27,405 bytes, SHA256: 1F94752EAEFA683AFF69E12554130D8D872C0245EB6A 4F46D4CB27B07D36988F
Backup method	pg_dump custom format (-Fc), compressed binary dump (2026-11-04)
Restore test result	Success (Restored into canteen_db_test_restore; all 7 tables, constraints, and data verified)

Access control after restore	Yes (Activity 4 roles and permissions validated; canteen_app DELETE restrictions enforced)
Recovery simulation result	Recovered (Simulated deleted test record successfully restored from verified backup state)
Column Header	Value
Proposed RPO and RTO	RPO: 15 minutes, RTO: 30 minutes
Evidence folder link	https://github.com/Nycri/brewpos/tree/main/activity-6/Evi de nce-act6
Feedback sent to owners	(04/11/2026)
PDF progress document	Activity6_Progress_Report_Group8.pdf
Last updated	2026-11-04

AI USAGE DECLARATION

Google Gemini 3.8 Flash was utilized during project evaluation to inspect database schema definitions, evaluate relational hierarchy structures, formulate role verification test scenarios, analyze database maintenance operations and query performance execution plans, verify backup creation and restoration procedures, and structure technical report documentation. All database statements, configuration checks, and verification procedures were directly executed and validated against PostgreSQL 18.6 within the local database environment.